Yash Kumar Singh

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Summary

AI Engineer with **1.6 years** of experience building **production GenAI systems** and **Python backend services**. Shipped LLM microservices, **RAG/GraphRAG** retrieval and asynchronous AI pipelines for international clients, and built **multi-agent** systems with LangGraph, A2A and MCP. Strong in FastAPI, PostgreSQL, Redis, Docker and Azure OpenAI, with a focus on **LLM evaluation**, guardrails and observability to ship reliable AI to production.

Experience

IQ Innovation Hub LLP **May 2025 – Present**
Software Engineer *Pune, India*

- **Enterprise RAG & GenAI Platform**: Built and deployed production **RAG systems**, LLM-powered microservices, and AI APIs using **Python**, **FastAPI**, **Azure OpenAI (GPT-4o)**, and **Anthropic Claude**, enabling structured task generation across **8 report types** for **7 properties**.
- **AI Data Pipelines**: Engineered asynchronous AI ingestion and generation workflows using **FastAPI**, **Celery**, and **Redis** for LLM orchestration and background processing, executing **56 active prompt templates** against production LLM APIs during daily batch runs.
- **Knowledge Retrieval**: Contributed to **hybrid retrieval** pipelines combining vector search and knowledge graphs (**Qdrant**, **Neo4j**) for a **GraphRAG**-based institutional-memory system answering natural-language queries.
- **Agentic AI Workflows**: Developed **agentic AI workflows** using **LangGraph** in internal POCs for stateful LLM orchestration, integrating retrieval, structured outputs, and controlled execution flows for AI-driven tasks.
- **AI Evaluation & Guardrails**: Enforced strict **JSON-schema** outputs across 4 LLM providers, added input sanitization and step-wise prompt checks to reduce false signals, logged per-run **tokens, latency and payloads** for evaluation, and contributed to code-side **guardrails** filtering hallucinated findings.
- **Backend & Security**: Implemented **OAuth2/JWT** authentication with **multi-tenant**, role-based access control across platform services, built **Microsoft Graph** webhook pipelines ingesting reports into **Azure Data Lake**, and added LLM usage logging and **rate limiting** on LLM endpoints to cap API costs.
- **Insurance Backend Platform**: Contributed to a **Spring Boot** insurance-aggregation platform for a **German client**, building **REST APIs** across **5+ external providers** and cutting end-to-end API latency by **98%**.

Projects

VoxDesk | *Python, LangGraph, A2A, MCP, FastAPI, PostgreSQL, Next.js* **Link**

- Built a **multi-agent voice AI platform** using **LangGraph**: a supervisor agent delegates to **RAG** and booking agents via the **A2A protocol**, with tools served by a custom **MCP server** and **HITL** approval.
- Developed a real-time **voice pipeline** with **speech-to-text** (Whisper), streaming LLM responses and **text-to-speech** (Kokoro, ElevenLabs) over **WebSockets**, supporting spoken confirmations and interruptions.
- Built an **LLM evaluation** suite with **LLM-as-a-judge**, reaching **86% Hit@1**, **100% intent routing** and **5/5 faithfulness**; added prompt-injection **guardrails**, LLM failover, rate limiting and **Langfuse** observability.

FilingLens | *Python, LangGraph, Qdrant, FastAPI, Gemini, React* **Link**

- Built a **RAG** system answering financial questions from **SEC 10-K filings** with cited answers, using **table-aware chunking** and a **LangGraph** pipeline with a zero-cost scope **guardrail** and **Gemini 2.5**.
- Implemented **hybrid retrieval** (dense + BM25 fused via RRF) with cross-encoder **reranking**, improving **Hit Rate@5 from 70% to 80%** on a hand-verified golden set.

Technical Skills

Languages:	Python, Java, C++, SQL, JavaScript
GenAI / Agents:	LangChain, LangGraph, CrewAI, MCP, A2A, Prompt Engineering, RAG, Langfuse
AI / ML:	AI Evaluation, NLP, Embeddings, Speech (STT/TTS)
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Vector / Graph DBs:	Qdrant, pgvector, FAISS, Neo4j
Frameworks:	FastAPI, Spring Boot, Next.js, Celery, Pydantic, REST APIs, GraphQL, gRPC, WebSockets
Cloud / DevOps:	Azure (Blob Storage, Pipelines), AWS, Docker, CI/CD
Developer Tools:	Git, GitHub, SQLAlchemy 2.0, OpenAI Agent SDK, Alembic, Postman, Swagger, Pytest

Education

Centre for Development of Advanced Computing (CDAC ACTS) **Aug 2024 – Feb 2025**
Post Graduate Diploma - Advance Computing *Pune, India*

KIET Group Of Institution, Delhi NCR (AKTU) **Sep 2020 – Jun 2024**
Bachelor of Technology in Information Technology *Delhi NCR, India*